SPA617 Radio Astronomy Laboratory

RF Components: Amplifier

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12th August 2026

1 Aim

To experimentally understand the operations of amplifier, it harmonics and third-order intercept.

2 Objectives

- To determine the gain of the amplifier.
- To observe and understand the harmonics obtained.
- To find the third-order intercept.

3 Theory

3.1 RF Amplifier Gain and Harmonics

The amplifier is used as a part of RF chain. Here the amplifier is used to amplify the weak signals received through the radio antenna. It amplifies the power of the signal and it is given as

$$G = P_{\text{out}} - P_{\text{in}}, \tag{1}$$

where P_{in} and P_{out} are the input and output powers, respectively, expressed in dBm.

An ideal amplifier would amplify a signal without introducing distortion. However, a practical amplifier becomes nonlinear at higher input powers. Its behavior can be represented by the polynomial

$$y(t) = a_1 x(t) + a_2 x^2(t) + a_3 x^3(t) + \cdots . \tag{2}$$

For a sinusoidal input

$$x(t) = A\cos(\omega t), \tag{3}$$

the output contains the fundamental frequency as well as harmonics. Considering the first three terms,

$$y(t) = a_1 A\cos(\omega t) + a_2 A^2 \cos^2(\omega t) + a_3 A^3 \cos^3(\omega t). \tag{4}$$

Using the trigonometric identities for $\cos^2(\omega t)$ and $\cos^3(\omega t)$, the output contains a DC component and frequency components at ω, 2ω, and 3ω:

$$y(t) = \frac{a_2 A^2}{2} + \left(a_1 A + \frac{3a_3 A^3}{4} \right) \cos(\omega t) + \frac{a_2 A^2}{2} \cos(2\omega t) + \frac{a_3 A^3}{4} \cos(3\omega t) + \cdots . \tag{5}$$

Thus, the nonlinear terms in the amplifier generate harmonics of the fundamental frequency.

At sufficiently high input power, the gain of the amplifier starts to decrease. The input power at which the gain falls by 1 dB from the small-signal gain is called the **1-dB compression point**. The corresponding output power is also specified as part of the compression-point measurement.

3.2 Third-Order Intercept (TOI)

Two frequencies f_1 and f_2 from two different sources are combined using a mixer and supplied to a non-liner amplifier. The out put contains the fundamental frequencies f_1 and f_2.harmonics, and third-order inter-modulation products. The two third-order products closest to the fundamentals are

$$2f_1 - f_2 \tag{6}$$

and

$$2f_2 - f_1. \tag{7}$$

The fundamental output increases with a slope of approximately $1:1$, while the third-order inter-modulation products output increases with a slope of approximately $3:1$. Their extrapolated intersection is called the third-order intercept (TOI)

4 Components Required

- RF signal generator(s)
- Spectrum analyzer
- RF amplifier
- RF mixer
- $50\,\Omega$ terminations
- Suitable RF cables

5 Experimental Setup

The RF signal generator is connected to the amplifier, and the output is measured using a spectrum analyzer to determine the amplifier gain.

For third order intercept (TOI), we use to two RF sources with frequencies f_1 and f_2 with same power is mixed using mixer. The output is connected to the amplifier and further connected to the spectrum analyzer.

6 Observations and Graphs

6.1 Amplifier Gain as a Function of Frequency

The input power was maintained at $-10\,\mathrm{dBm}$, and the output power was measured at different frequencies.

Table 1: Measured gain as a function of frequency.

First Half				Second Half			
Frequency (MHz)	P_{in} (d)	P_{out} (d)	Gain (dB)	Frequency (MHz)	P_{in} (d)	P_{out} (d)	Gain (dB)
50	-10	13.81	23.81	1150	-10	11.62	21.62
150	-10	13.28	23.28	1250	-10	11.35	21.35
250	-10	12.92	22.92	1350	-10	11.13	21.13
350	-10	12.62	22.62	1450	-10	10.92	20.92
450	-10	12.52	22.52	1550	-10	10.47	20.47
550	-10	12.20	22.20	1650	-10	10.30	20.30
650	-10	12.23	22.23	1750	-10	10.10	20.10
750	-10	12.12	22.12	1850	-10	9.53	19.53
850	-10	11.87	21.87	1950	-10	8.40	18.40
950	-10	11.60	21.60	2000	-10	8.20	18.20
1050	-10	11.63	21.63				

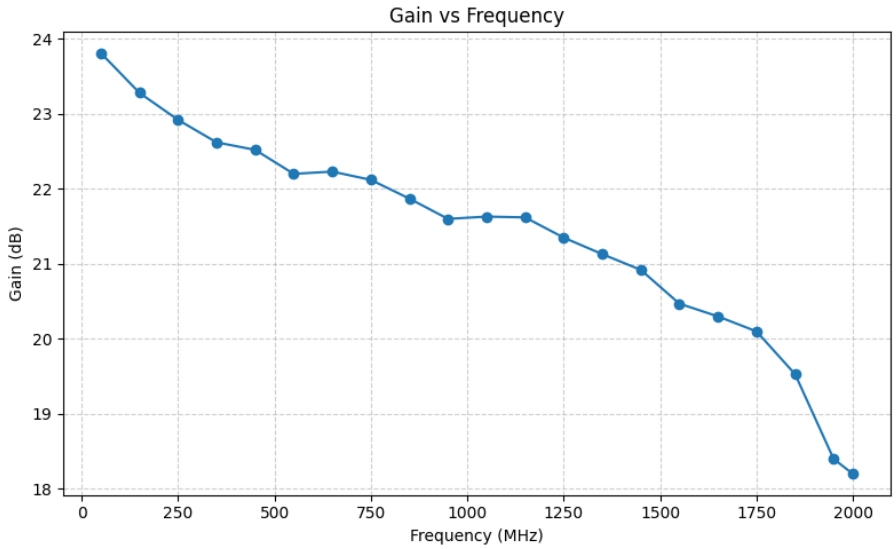


Figure 1: Measured gain of the amplifier as a function of frequency.

6.2 Harmonics and 1-dB Compression Point

The frequency applied to the amplifier was fixed at

$$f = 50\,\mathrm{MHz}. \tag{8}$$

The input power was varied and the corresponding output power was measured.

Table 2: Input-output power characteristics of the nonlinear amplifier.

P_{in} (d)	P_{out} (d)	Gain (dB)
-20	3.80	23.80
-19	4.80	23.80
-18	5.81	23.81
-17	6.80	23.80
-16	7.76	23.76
-15	8.76	23.76
-14	9.75	23.75
-13	10.74	23.74
-12	11.70	23.70
-11	12.69	23.69
-10	13.62	23.62
-9	14.50	23.50
-8	15.47	23.47
-7	16.25	23.25
-6	16.85	22.85
-5	17.29	22.29

The corresponding plots of output power and gain as a function of input power are shown in Figures 2 and 3.

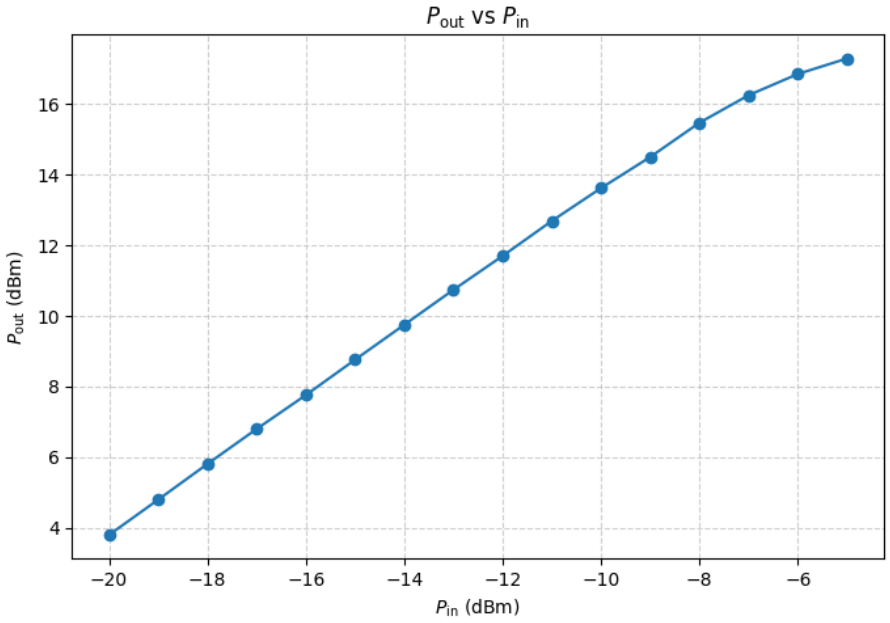


Figure 2: Output power P_{out} as a function of input power P_{in}.

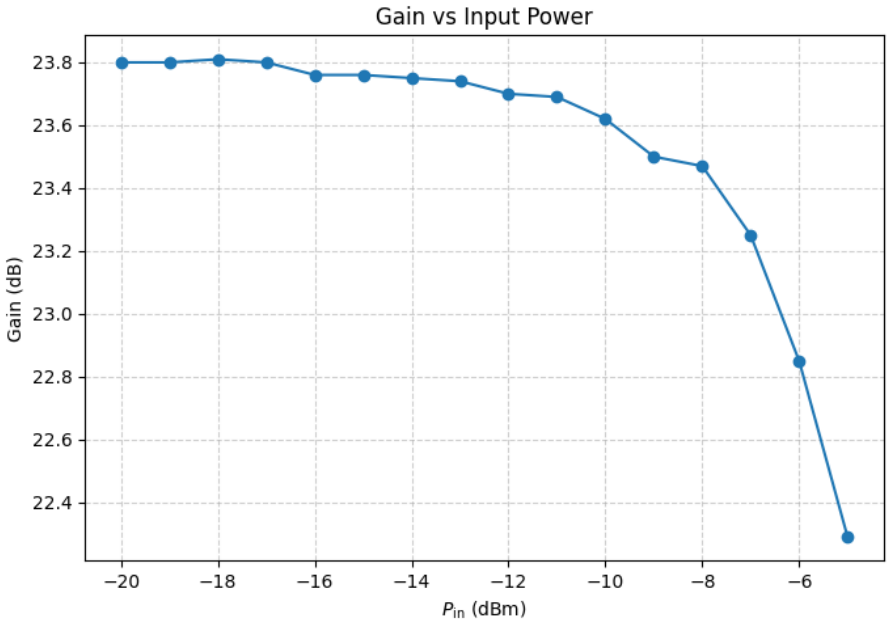


Figure 3: Amplifier gain as a function of input power.

From the gain-versus-input-power plot, the amplifier exhibits nearly constant gain at low input powers. As the input power increases, the gain starts decreasing, indicating the onset of nonlinear operation

6.3 Third-Order Intercept (TOI)

The two input frequencies were

$$f_1 = 13\,\text{MHz}, \qquad f_2 = 10\,\text{MHz}. \tag{9}$$

The two third-order intermodulation products are

$$2f_1 - f_2 = 2(13) - 10 = 16\,\text{MHz}, \tag{10}$$
$$2f_2 - f_1 = 2(10) - 13 = 7\,\text{MHz}. \tag{11}$$

The input powers of the two signals were

$$P_1 = P_2 = -10\,\text{dBm}. \tag{12}$$

Table 3: Third-order intermodulation measurement data.

P_{in} (d)	P_{out} (d)	P_{3rd} (d)
-10	10.36	-25.80
-9	11.36	-21.00
-8	12.24	-18.20
-7	13.20	-15.21
-6	14.00	-12.60
-5	14.80	-10.00

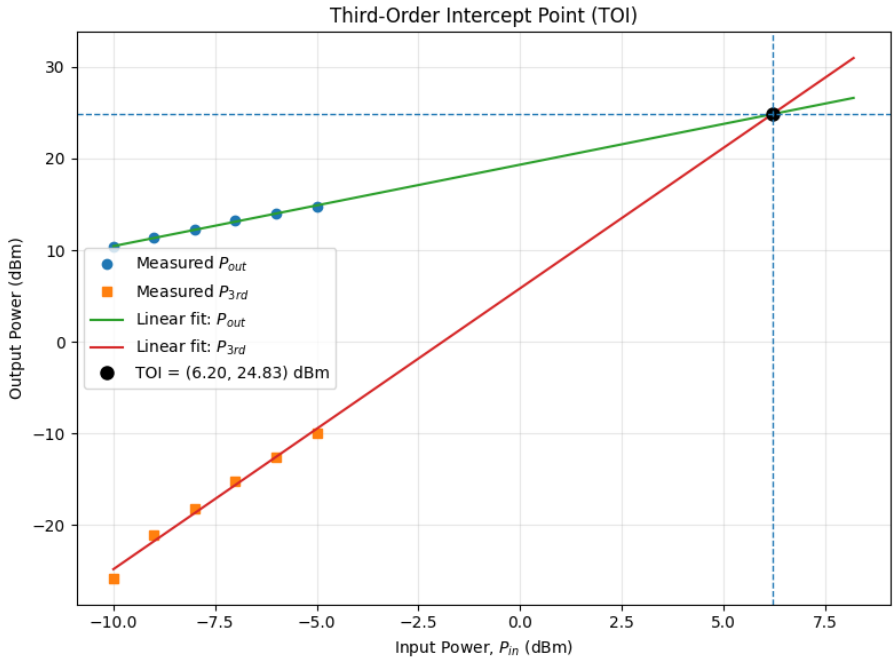


Figure 4: Fundamental and third-order intermodulation output power as a function of input power.

From the linear fits to the measured data, the slopes of the fundamental and third-order product curves are approximately

$$m_{\text{fundamental}} \approx 0.88 \tag{13}$$

and

$$m_{\text{3rd}} \approx 3.01. \tag{14}$$

The theoretical slopes are approximately $1:1$ and $3:1$, respectively. For the measured data, the fitted equations are approximately

$$P_{\text{out}} = 0.888 P_{\text{in}} + 19.32, \tag{15}$$
$$P_{\text{3rd}} = 3.063 P_{\text{in}} + 5.834. \tag{16}$$

By equating the two fitted lines:

$$0.888 P_{\text{in}} + 19.32 = 3.063 P_{\text{in}} + 5.834. \tag{17}$$

Therefore,

$$P_{\text{in,TOI}} \approx 6.20\,\text{dBm}. \tag{18}$$

and

$$P_{\text{out,TOI}} \approx 24.83\,\text{dBm}. \tag{19}$$

7 Results

The important measured and calculated results are summarized in Table 4.

Table 4: Summary of measured and calculated results.

Quantity	Symbol	Measured/Calculated Value
Amplifier input power	P_{in}	$-10\,\text{dBm}$
Amplifier output power	P_{out}	$13.81\,\text{dBm}$
Amplifier gain	G	$23.81\,\text{dB}$
1-dB compression point	P_{in}	$-6\,\text{dBm}$
Third-order intercept point	$P_{\text{in,TOI}}$	$6.20\,\text{dBm}$
Third-order intercept output	$P_{\text{out,TOI}}$	$24.83\,\text{dBm}$

8 Conclusion

Thus we have experimentally analyzed the operations of amplifier, its harmonies and third order intercepts

- The measured gain decreases with increasing frequency, impling that the amplifier gain is dependent on frequency
- At low input powers, the amplifier gives an approximately constant gain.
- The harmonic generation observed in the output shows the nonlinear behavior of the amplifier.

References

[1] SPA617 Radio Astronomy Laboratory, *RF Components: Characterization and Measurement*

AI Usage

ChatGPT was used for plotting the experimental data and formatting the report in LaTeX.